# DOKUMENTASI
# UJI KOMPETENSI KEAHLIAN
## PENGEMBANGAN SISTEM PARKIR PINTAR BERBASIS IOT


Disusun Oleh:

**Hisan Ramadhan Putra**

**NIS. 2324.10.360**

**XII RPL 1**


**BIDANG KEAHLIAN TEKNOLOGI INFORMASI DAN KOMUNIKASI**

**PROGRAM KEAHLIAN PENGEMBANGAN PERANGKAT LUNAK DAN GIM**

**KONSENTRASI KEAHLIAN REKAYASA PERANGKAT LUNAK**

**SMK MAHARDHIKA BATUJAJAR**

**2026**

**LAPORAN PENGERJAAN UJI KOMPETENSI KEAHLIAN**

**SMK MAHARDHIKA BATUJAJAR**

**TAHUN PEMBELAJARAN 2025 – 2026**

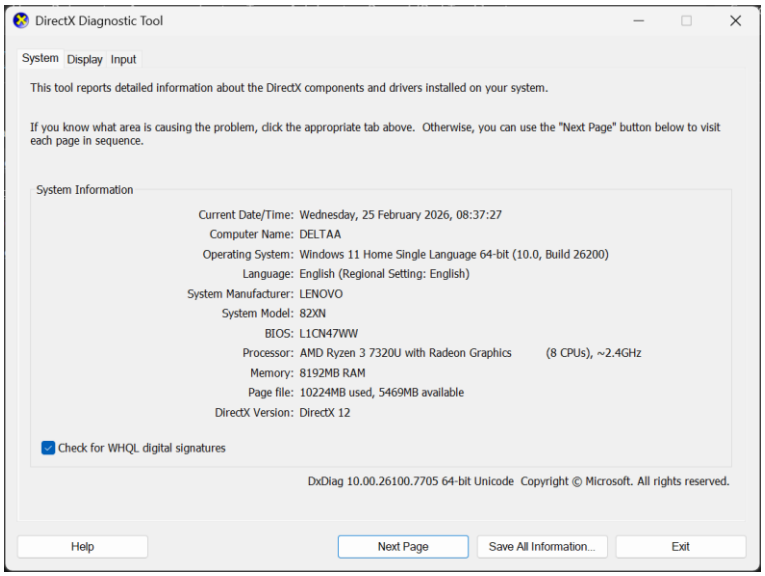
| | |
|---|---|
| Konsentrasi Keahlian | : Rekayasa Perangkat Lunak |
| Kelas | : XII RPL 1 |
| Nama Lengkap | : Hisan Ramadhan Putra |
| Nomor Induk Siswa | : 2324.10.360 |
| Judul Tugas | : Pengembangan Sistem Parkir Pintar Berbasis IoT |
| Jenis Aplikasi | : Situs Web/*Website* |
| Alokasi Waktu | : 12 Jam |

## 1. JADWAL KEGIATAN

| Kegiatan | Januari | | Februari | | | | | | | | | Mret | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 28 | 31 | 03 | 06 | 09 | 12 | 15 | 18 | 21 | 24 | 27 | 02 | 05 | 08 | 11 | 14 | 17 | 20 | 23 | 26 | 29 |
| Analisis Kebutuhan Sistem | | | | | | | | | | | | | | | | | | | | | |
| Pemodelan Aplikasi | | | | | | | | | | | | | | | | | | | | | |
| Perancangan Antarmuka | | | | | | | | | | | | | | | | | | | | | |
| Implementasi Aplikasi | | | | | | | | | | | | | | | | | | | | | |
| Pengujian | | | | | | | | | | | | | | | | | | | | | |
| Dokumentasi | | | | | | | | | | | | | | | | | | | | | |
| Pembuatan UG | | | | | | | | | | | | | | | | | | | | | |

## 2. PERSIAPAN PERANGKAT KERAS

### a. Laptop



b. *Microcontroller* ESP32

Berfungsi mengatur, mengoordinasikan, dan mengendalikan seluruh komponen yang digunakan dalam sistem, serta memproses *input* yang diterima dan mengirimkan *output* sebagai hasil dari proses tersebut.

c. MFRC522 RFID Reader dan Tag

Berfungsi mengirimkan data pengemudi kepada perangkat ESP lalu diproses lebih lanjut oleh ESP untuk menjalankan perintah tertentu.

d. Motor Servo

Berfungsi mengendalikan pergerakan portal berdasarkan perintah yang diterima dari ESP.

e. SSD1306 OLED Display

Berfungsi menampilkan pesan kepada pengemudi berdasarkan perintah yang dikirimkan oleh ESP.

3. PERSIAPAN PERANGKAT LUNAK

a. Astah

Digunakan dalam proses perancangan dan pemodelan sistem seperti *activity diagram* dan *usecase diagram*.

b. Balsamiq

Digunakan dalam proses perancangan antarmuka sistem untuk merancang tampilan visual, struktur navigasi, serta tata letak elemen.

c. Visual Studio Code

Digunakan dalam proses penulisan kode program sistem secara menyeluruh, mulai dari pengembangan antarmuka pengguna (*front-end*) hingga implementasi logika pemrograman (*back-end*).

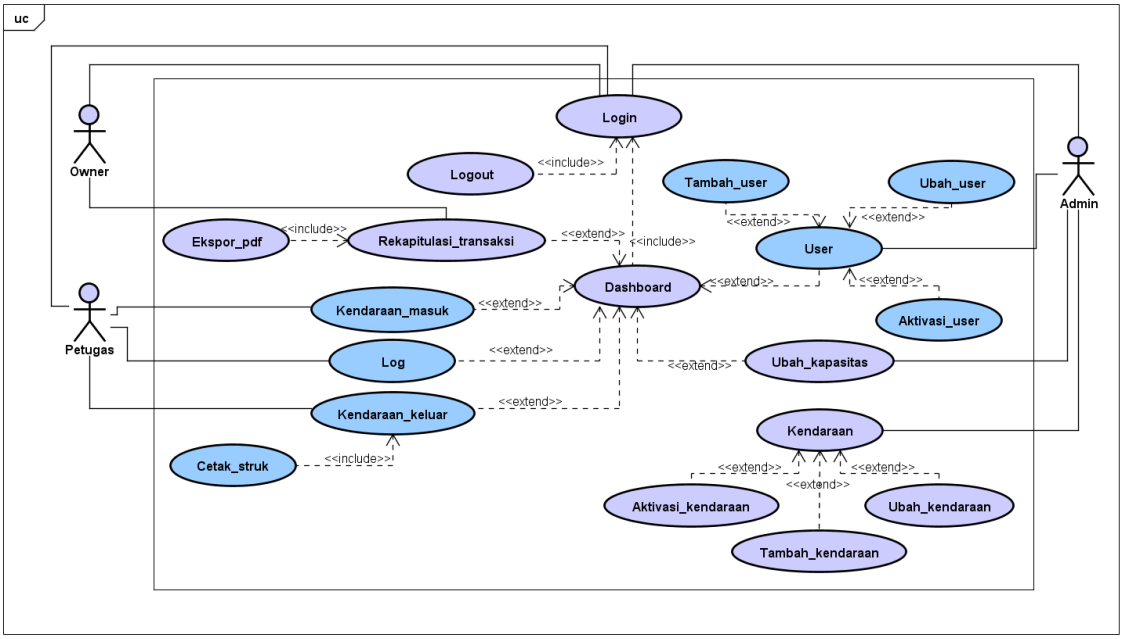
d. XAMPP

Digunakan sebagai server lokal untuk menjalankan MySQL sebagai DBMS, yang berfungsi menyimpan dan mengelola berbagai data penting seperti data *user*, transaksi, dan kendaraan.

e. Microsoft Word

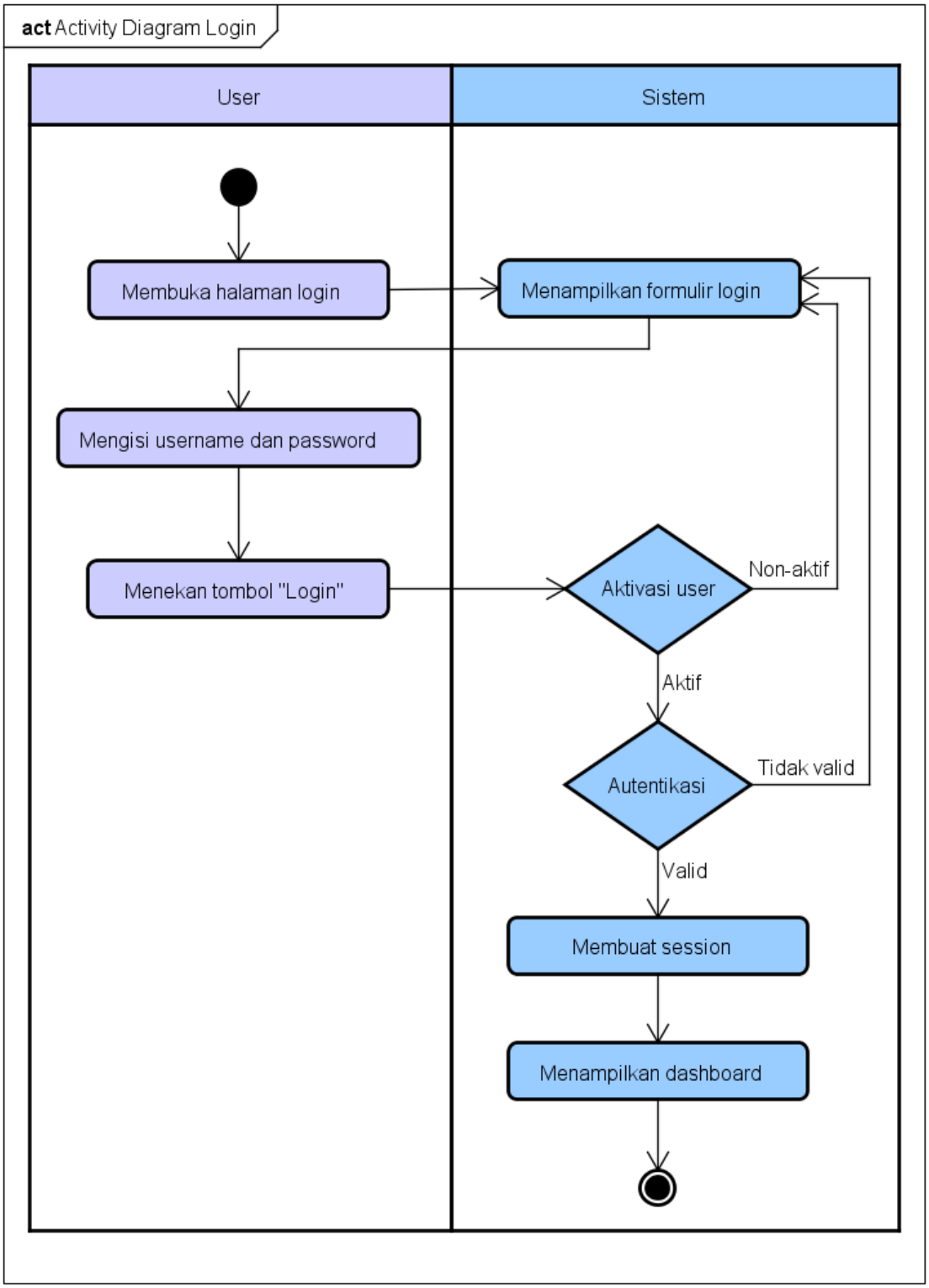
Digunakan dalam penyusunan dokumentasi sistem yang menjelaskan alur sistem serta penyusunan *user guide* (panduan pengguna) yang menjelaskan setiap fitur yang ada pada sistem.

# 4. STRUKTUR APLIKASI

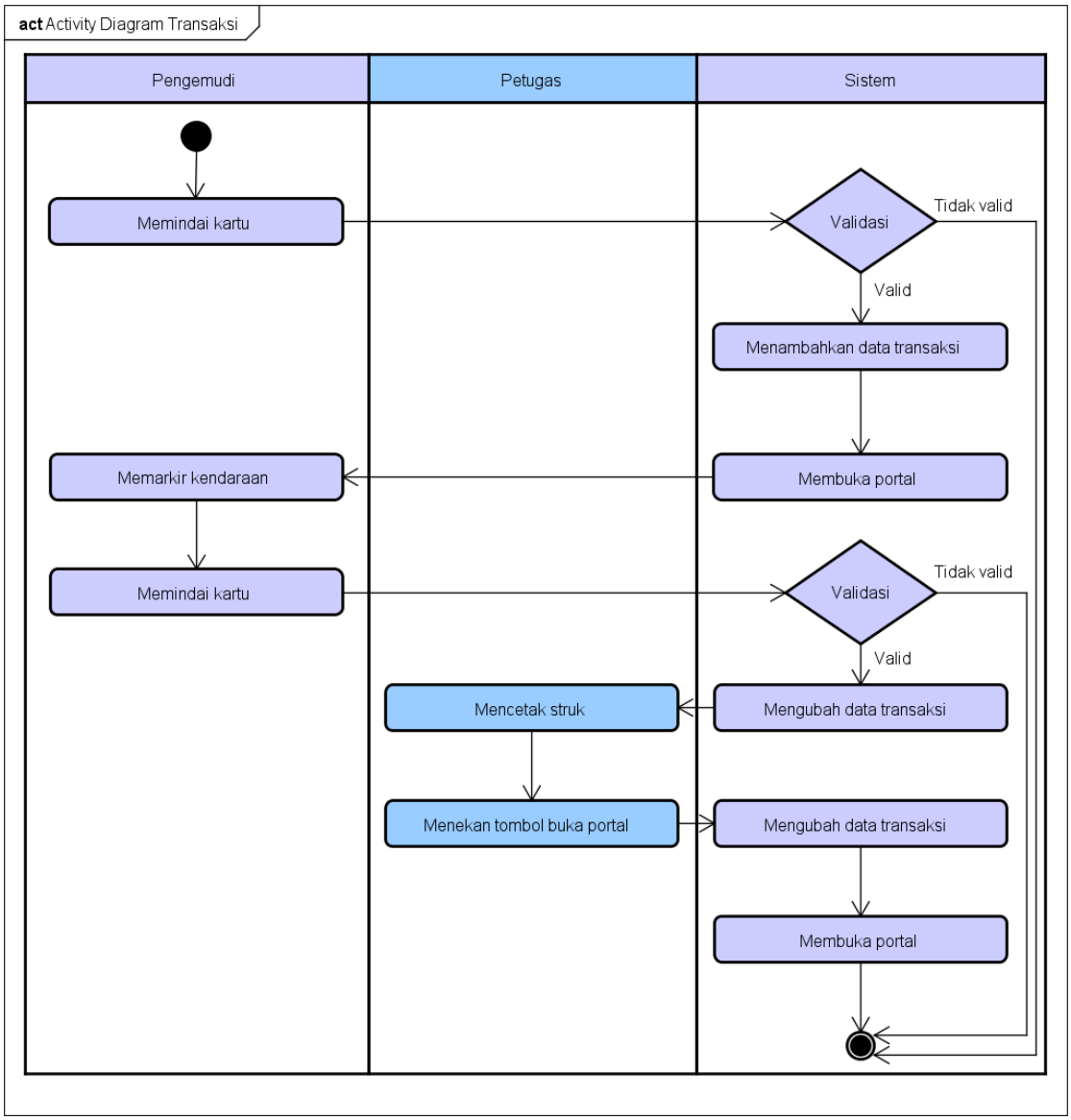


5. ALUR SISTEM

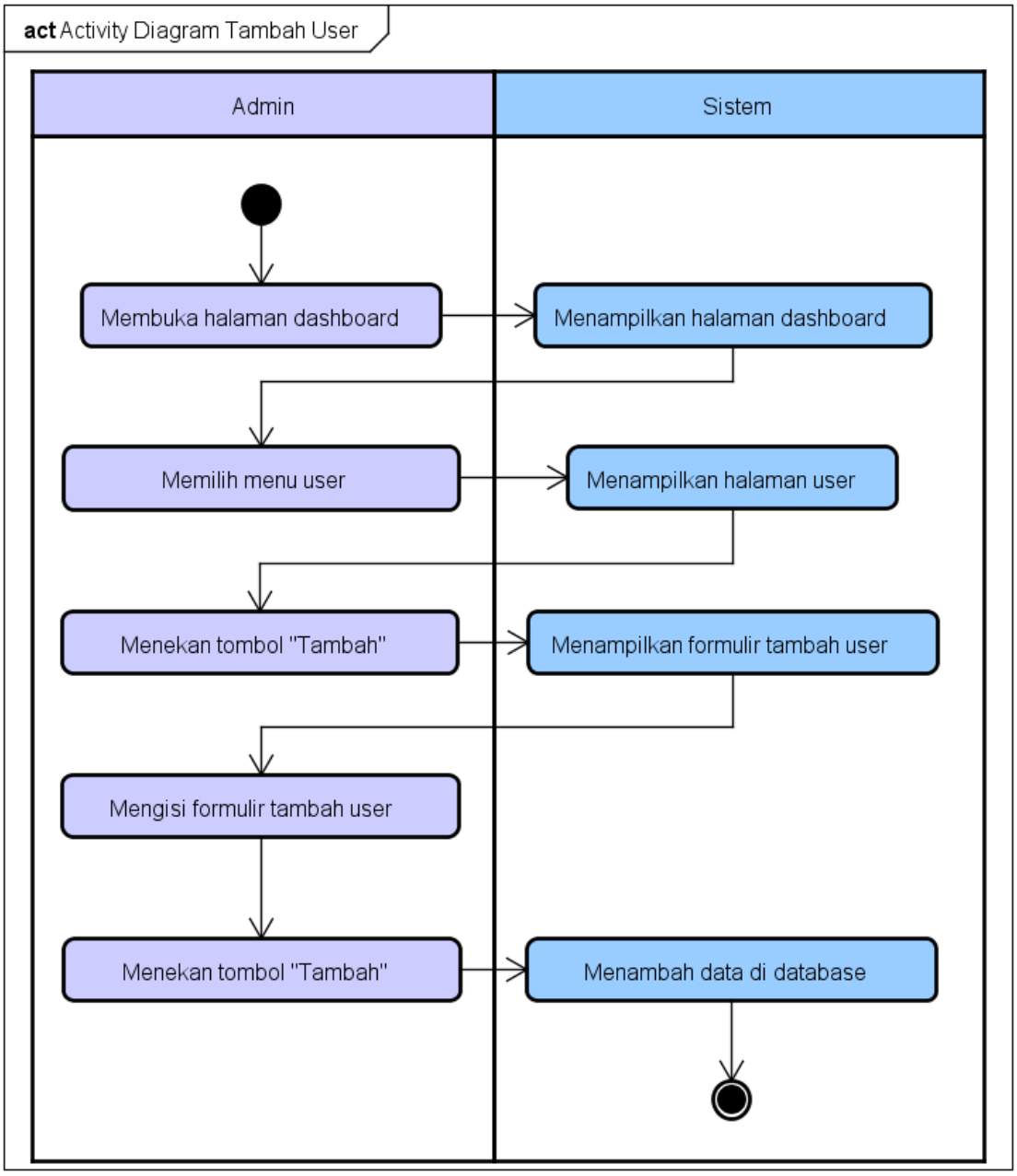
a. *Login*



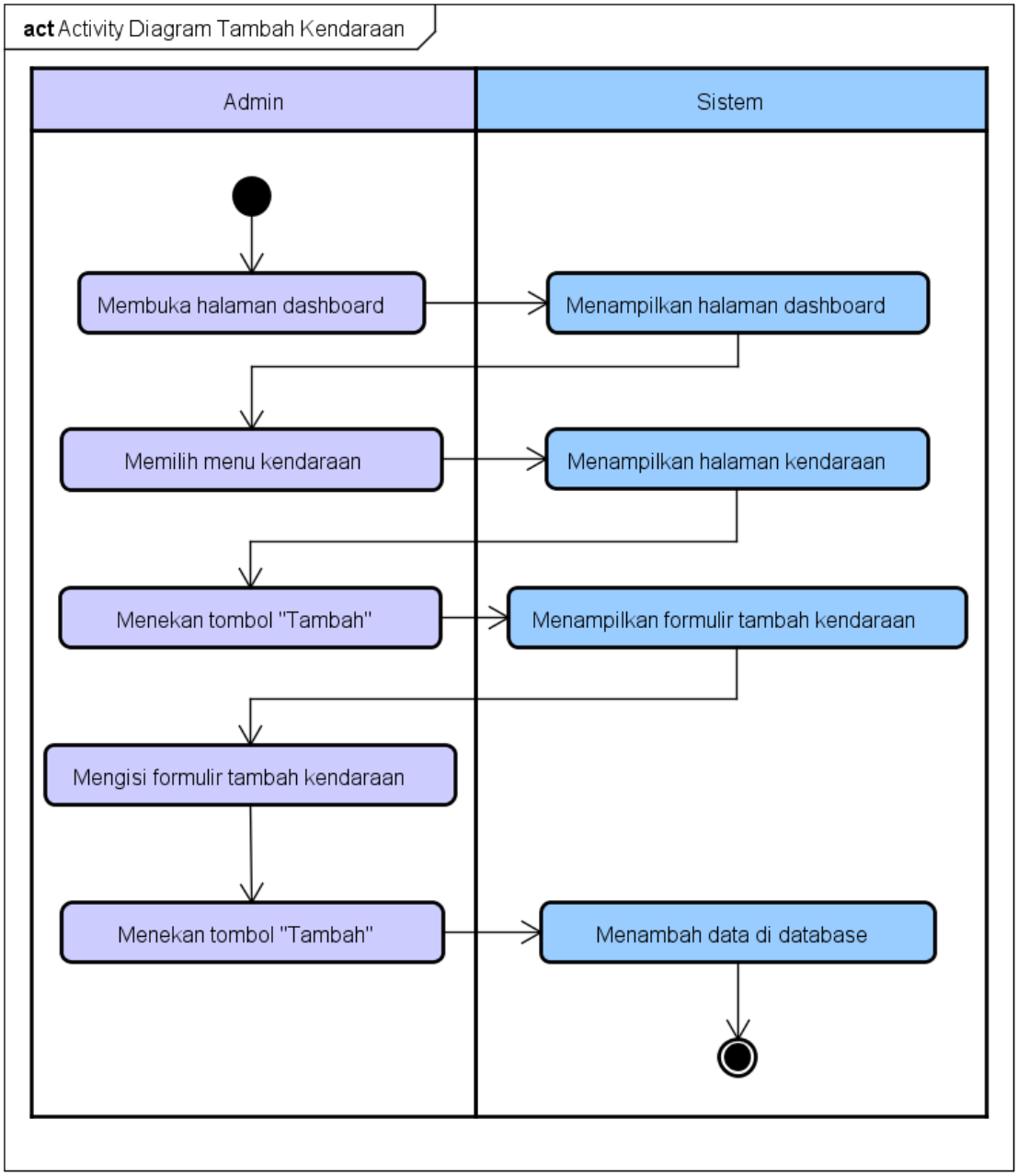
b. Transaksi


act Activity Diagram Transaksi
Pengemudi
Petugas
Sistem
Memindai kartu
Validasi
Tidak valid
Valid
Menambahkan data transaksi
Membuka portal
Memarkir kendaraan
Memindai kartu
Validasi
Tidak valid
Valid
Mencetak struk
Mengubah data transaksi
Menekan tombol buka portal
Mengubah data transaksi
Membuka portal

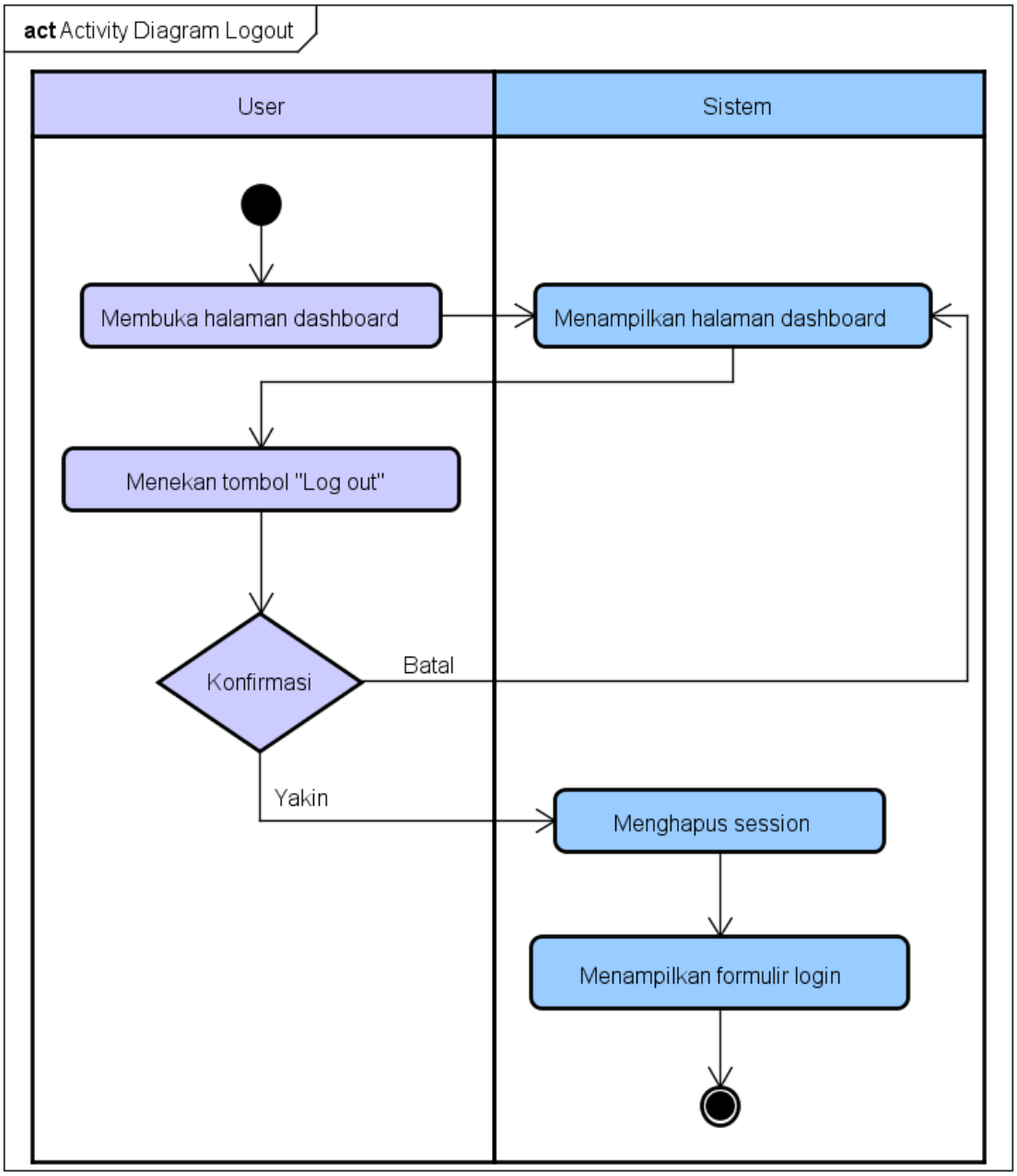
c. Tambah *User*



d. Tambah Kendaraan

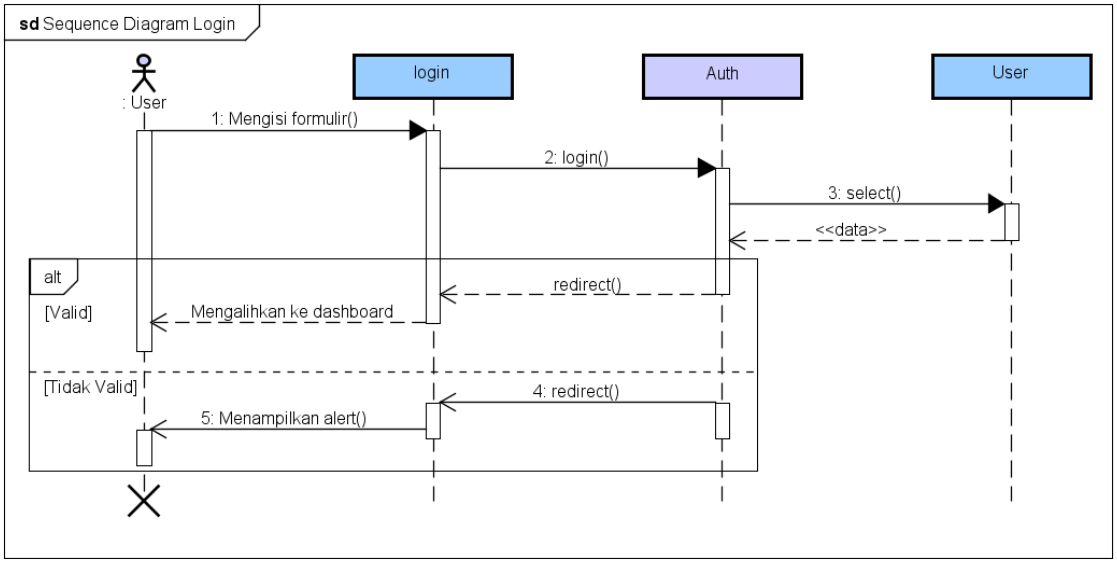


e. *Logout*

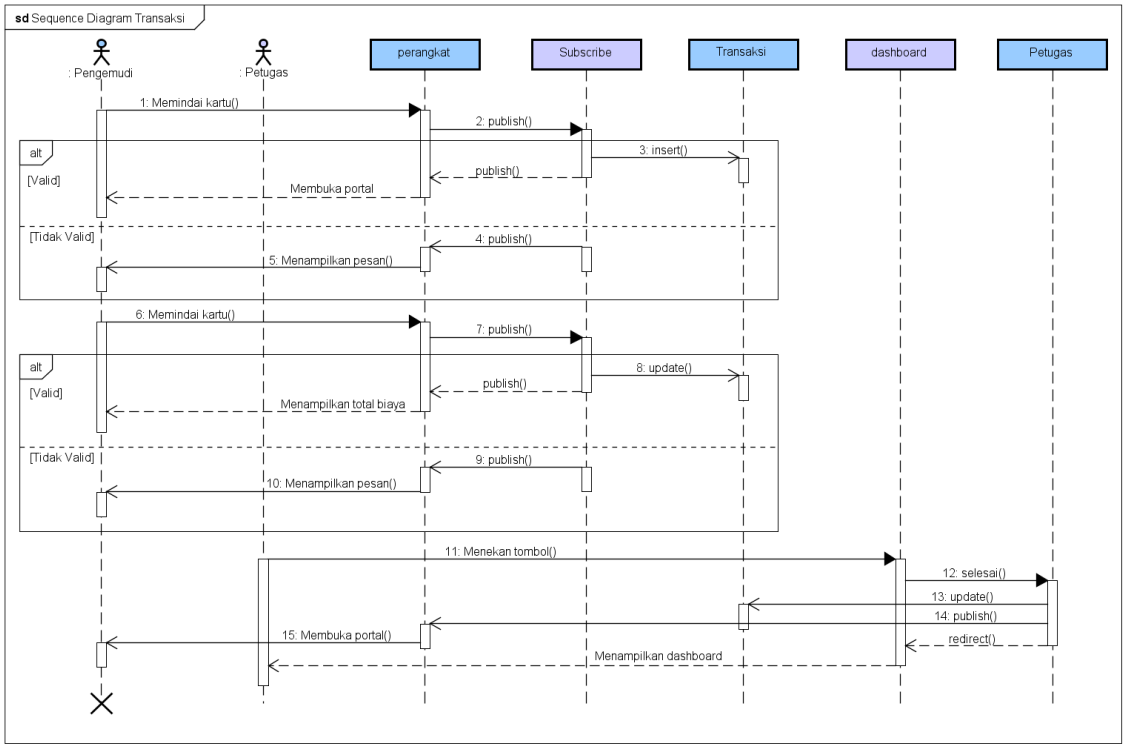

act Activity Diagram Logout
User
Sistem
Membuka halaman dashboard
Menampilkan halaman dashboard
Menekan tombol "Log out"
Konfirmasi
Batal
Yakin
Menghapus session
Menampilkan formulir login

## 6. ALUR INTERAKSI ANTAR OBYEK

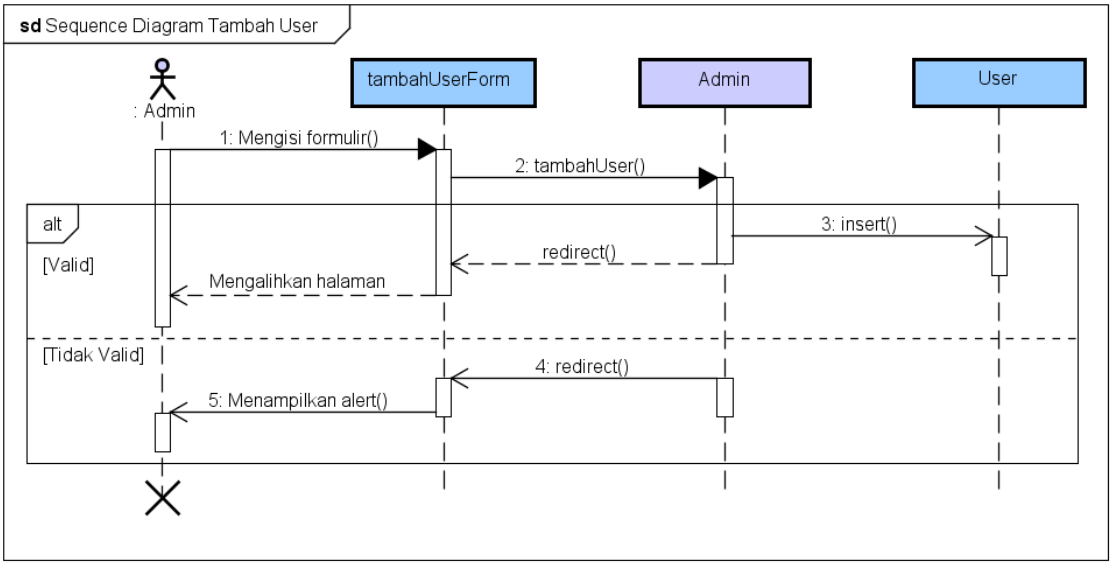
### a. *Login*



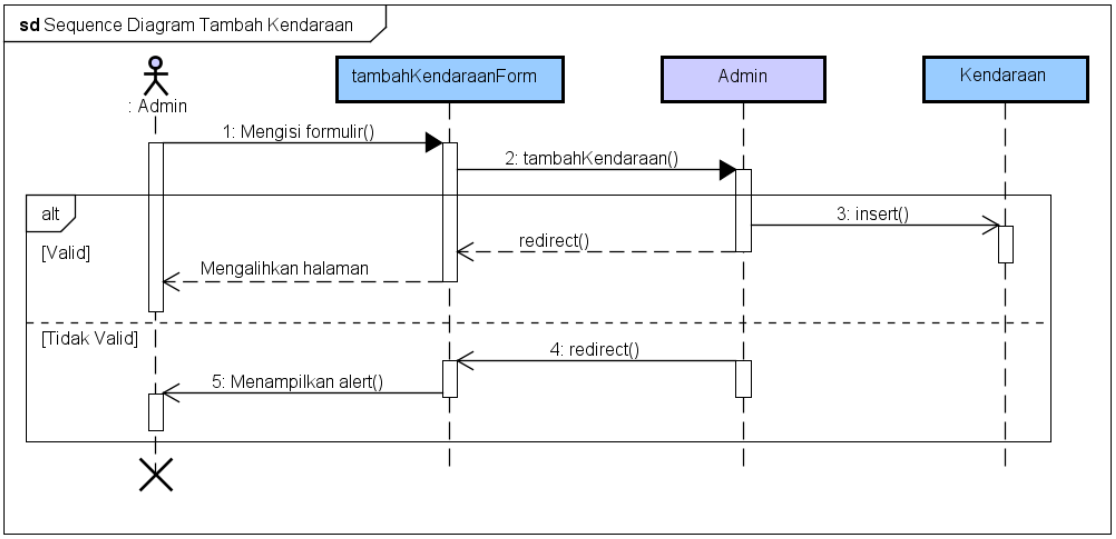

### b. Transaksi



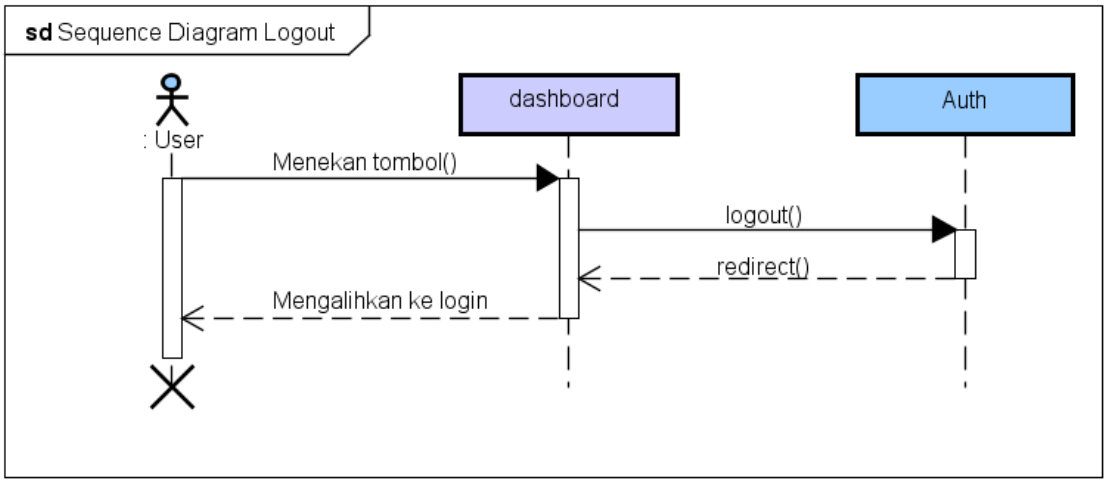
c. Tambah *User*



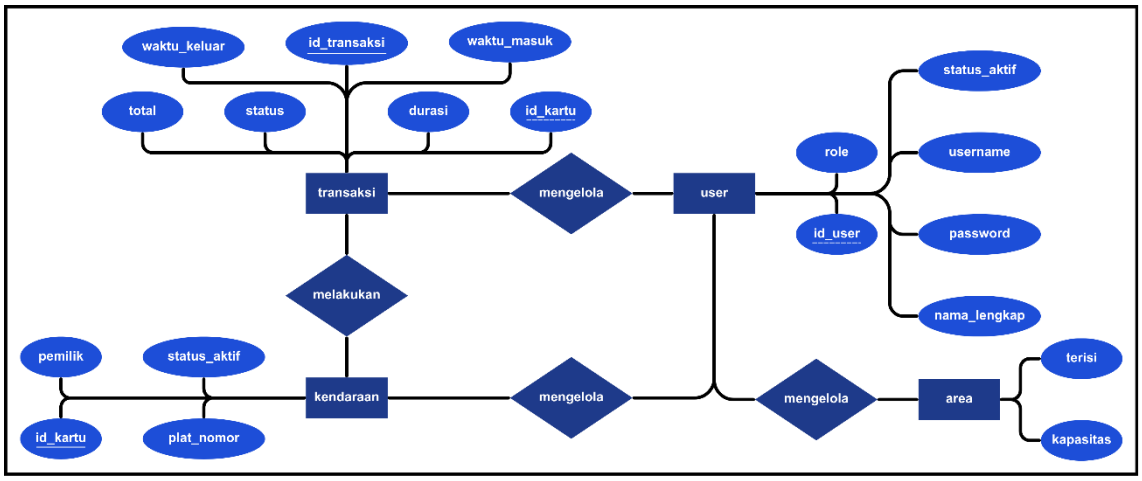

d. Tambah Kendaraan

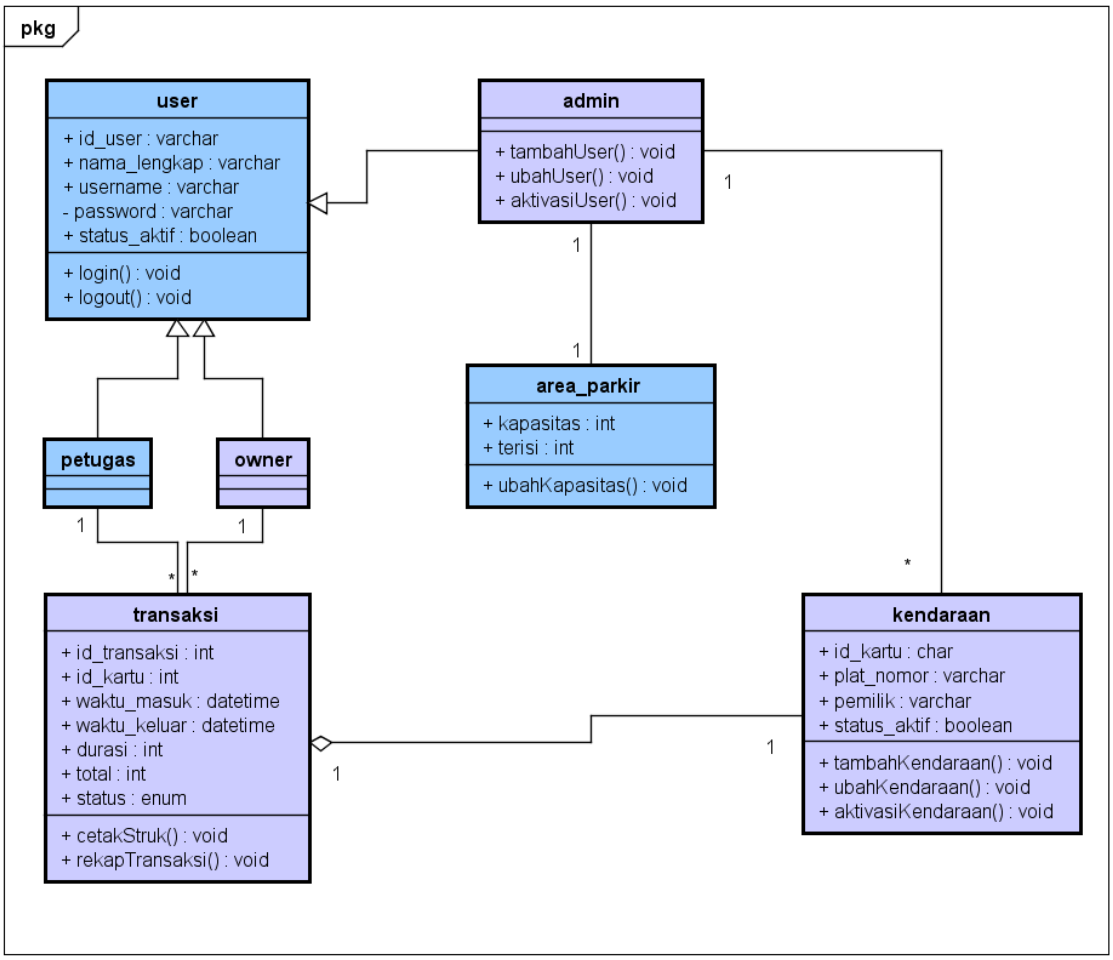



e. *Logout*



## 7. PERANCANGAN BASIS DATA



8. *SCENARIO USE CASE*

a. *Name* : *Login*

IDE : U-01

*Description* : Pengguna yang terdaftar di *database* dapat mengakses berbagai fitur dalam sistem berdasarkan *role*

*Actor(s)* : Petugas, Admin, Owner

| ***Main Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| 1 | Membuka halaman *login* | | |
| | | 2 | Menampilkan halaman *login* |
| 3 | Mengisi formulir *login* | | |
| 4 | Menekan tombol “Login” | | |
| | | 5 | Membuat *session* pengguna |
| | | 6 | Menampilkan halaman *dashboard* |

| ***Alternative Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

| ***Exception Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

b. *Name* : Transaksi

*IDE* : P-01

*Description* : Data kendaraan yang masuk dan keluar dicatat oleh sistem dan dikelola oleh petugas

*Actor(s)* : Petugas

| ***Main Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| 1 | Memindai kartu RFID | | |
| | | 2 | Menambahkan data transaksi |
| | | 3 | Menampilkan pesan di OLED |
| | | 4 | Membuka motor servo |
| 5 | Memarkir kendaraan | | |
| 6 | Memindai kartu RFID | | |
| | | 7 | Mengubah data transaksi |
| | | 8 | Menampilkan pesan di OLED |
| 9 | Menekan tombol “Cetak Struk” | | |
| | | 10 | Mengunduh struk transaksi |
| 11 | Menekan tombol “Buka Portal” | | |
| | | 12 | Mengubah data transaksi |
| | | 13 | Menampilkan pesan di OLED |
| | | 14 | Membuka motor servo |

| ***Alternative Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

| ***Exception Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

c. *Name* : Tambah *User*

*IDE* : A-01

*Description* : Penambahan pengguna sistem yang dapat mengakses fitur-fitur tertentu berdasarkan *role*

*Actor(s)* : Admin

| ***Main Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| 1 | Memilih menu “User” | | |
| | | 2 | Menampilkan halaman *user* |
| 3 | Menekan tombol “Tambah” | | |
| | | 4 | Menampilkan halaman formulir tambah *user* |
| 5 | Mengisi formulir tambah *user* | | |
| 6 | Menekan tombol ”Tambah” | | |
| | | 7 | Menambah data di *database* |

| ***Alternative Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

| ***Exception Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

d. *Name* : Tambah Kendaraan

*IDE* : A-02

*Description* : Penambahan kendaraan yang dapat memarkir di area parkir

*Actor(s)* : Admin

| ***Main Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| 1 | Memilih menu “Kendaraan” | | |
| | | 2 | Menampilkan halaman kendaraan |
| 3 | Menekan tombol “Tambah” | | |
| | | 4 | Menampilkan halaman formulir tambah *user* |
| 5 | Mengisi formulir tambah *user* | | |
| 6 | Menekan tombol ”Tambah” | | |
| | | 7 | Menambah data di *database* |

| ***Alternative Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

| ***Exception Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

e. *Name* : *Logout*
*IDE* : U-02
*Description* : Pengguna mengakhiri sesi *login*
*Actor(s)* : Petugas, Admin, Owner

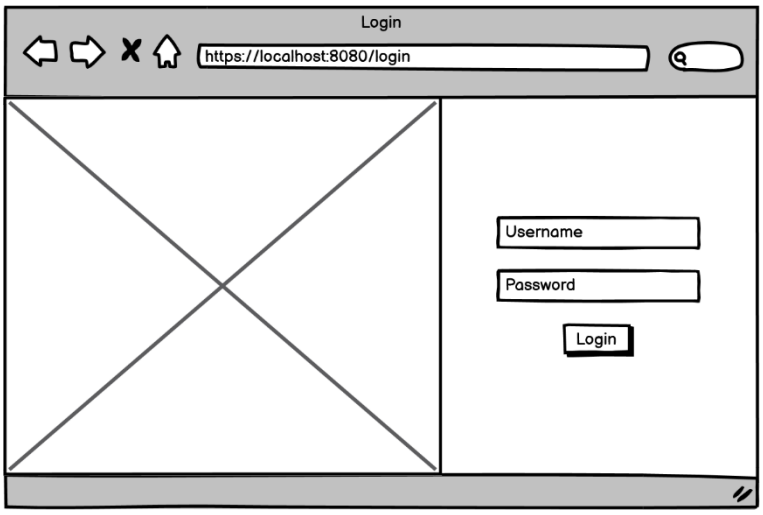
| ***Main Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| 1 | Menekan menu “Log out” | | |
| | | 2 | Menampilkan konfirmasi *log out* |
| 3 | Menekan tombol “Yakin” | | |
| | | 4 | Menghapus *session* pengguna |
| | | 5 | Menampilkan halaman *login* |

| ***Alternative Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

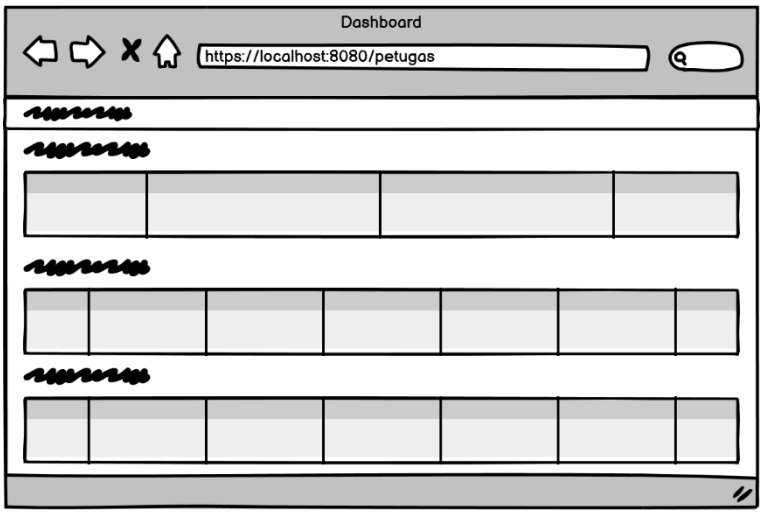
| ***Exception Course*** | | | |
|---|---|---|---|
| Aksi Aktor | | Reaksi Sistem | |
| | | | |
| | | | |
| | | | |

9. PERANCANGAN ANTARMUKA

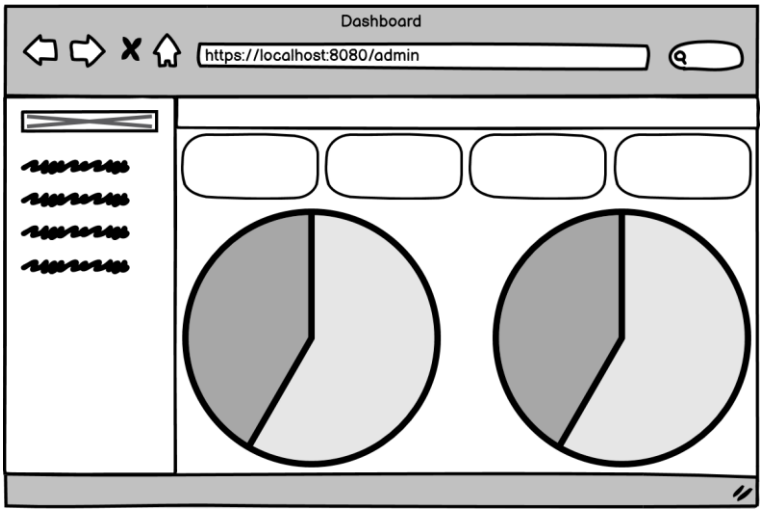
   a. *Login*



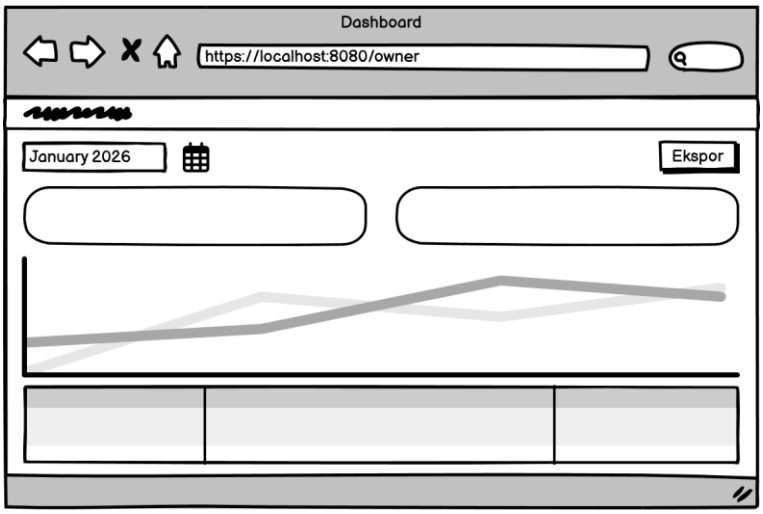
b. *Dashboard* Petugas



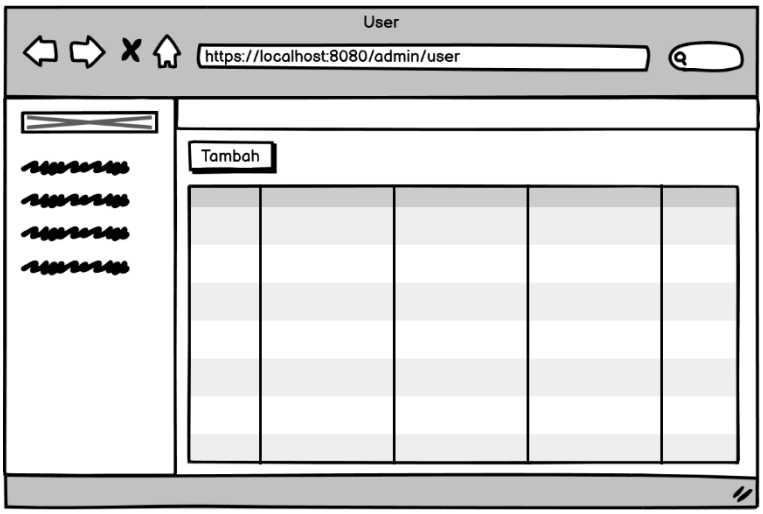

c. *Dashboard* Admin



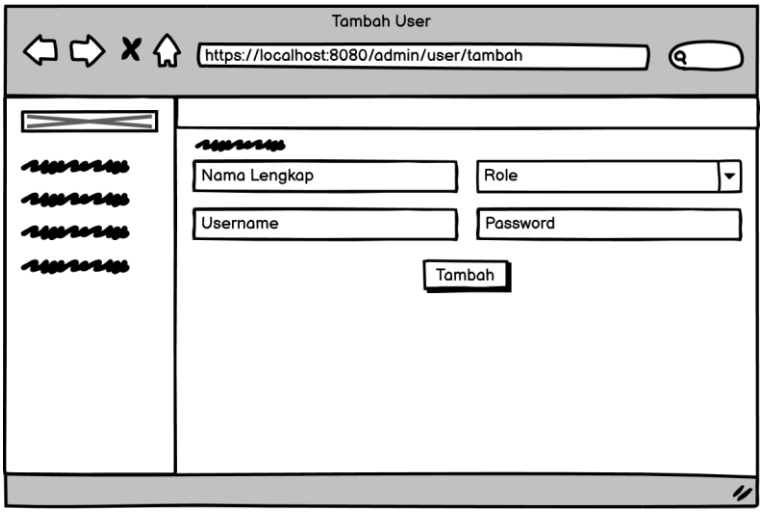

d. *Dashboard Owner*



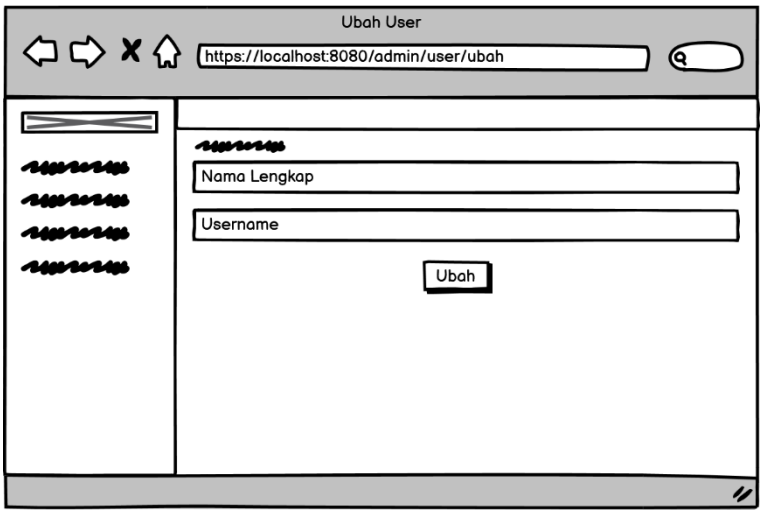
e. Daftar *User*



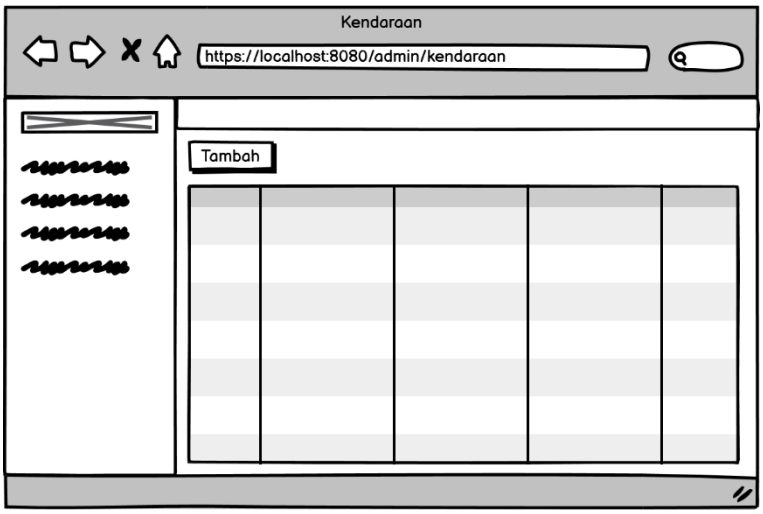

f. Formulir Tambah *User*



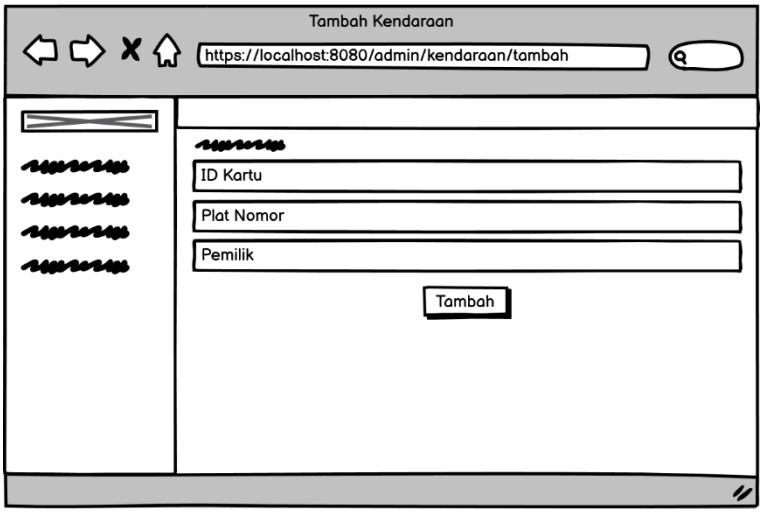

g. Formulir Ubah *User*



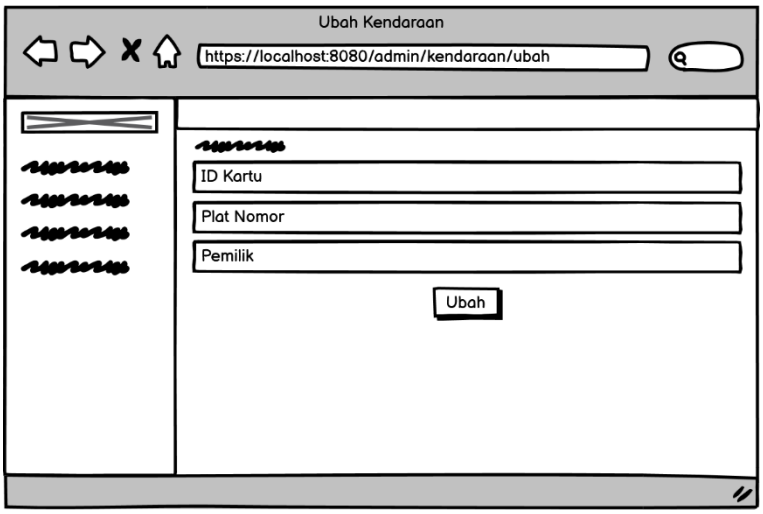
h. Daftar Kendaraan




i. Formulir Tambah Kendaraan

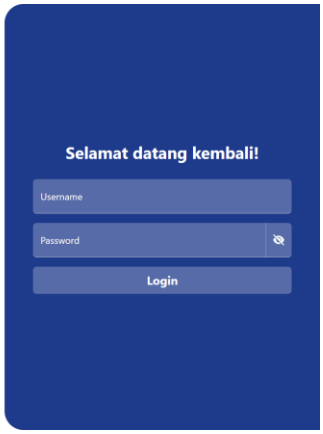



j. Formulir Ubah Kendaraan

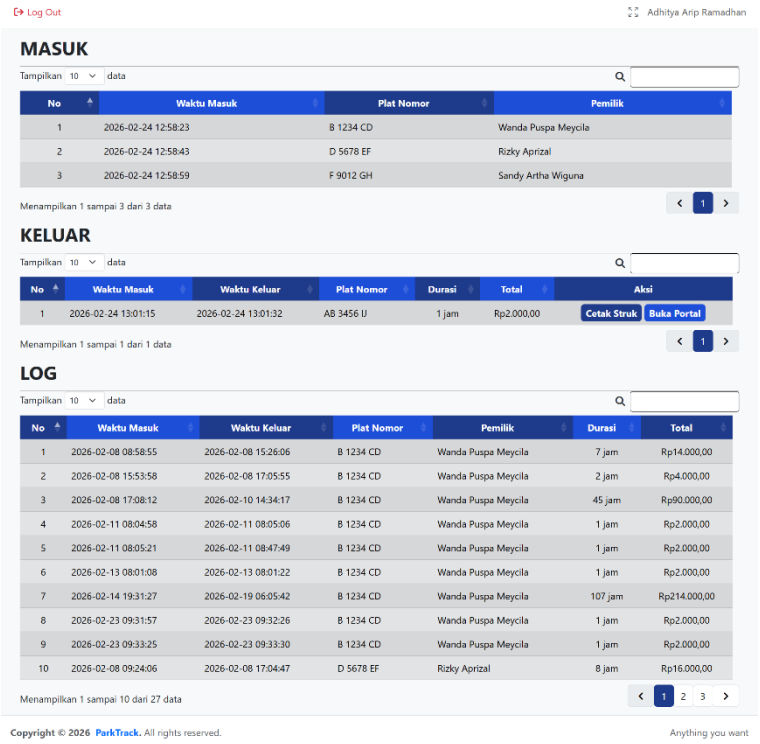


## 10. IMPLEMENTASI ANTARMUKA

### a. *Login*

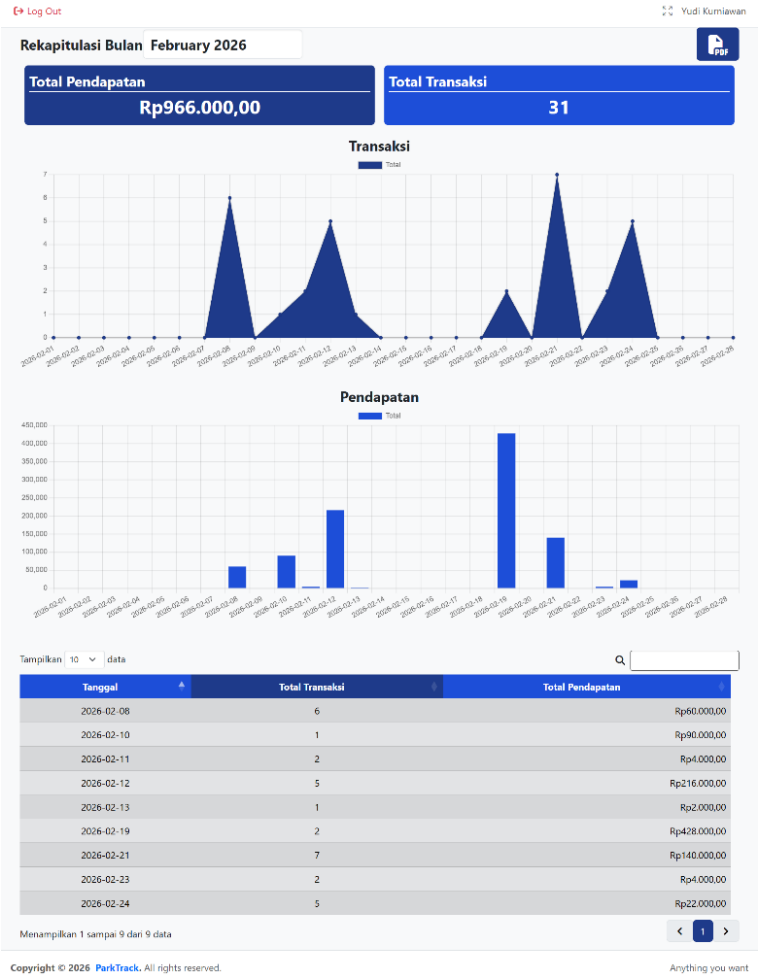



### b. *Dashboard* Petugas

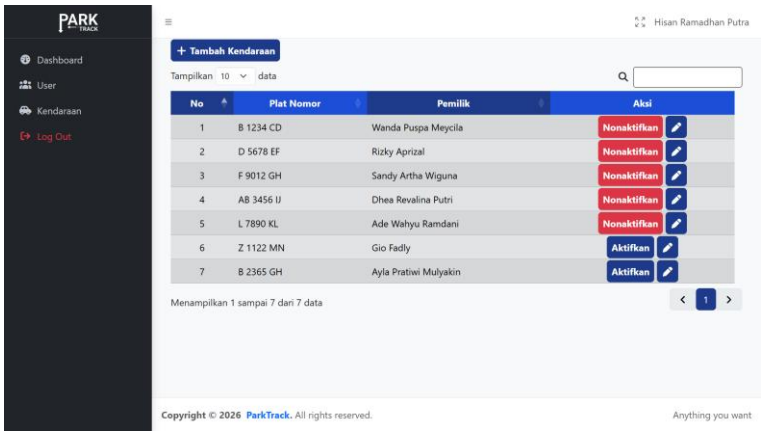


c. *Dashboard* Admin

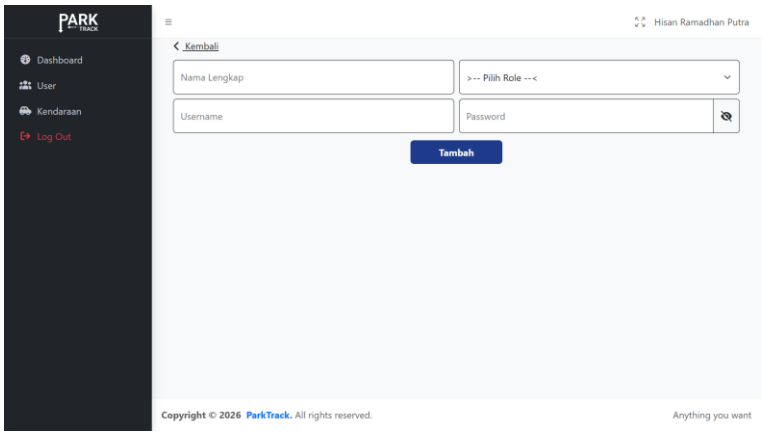
d. *Dashboard Owner*



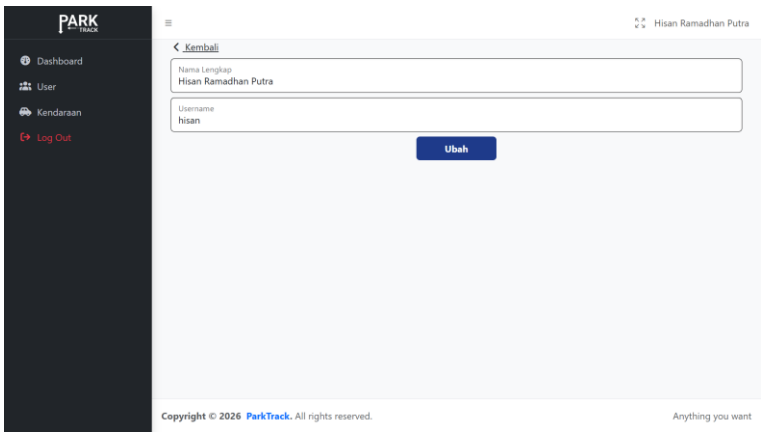
e. Daftar *User*



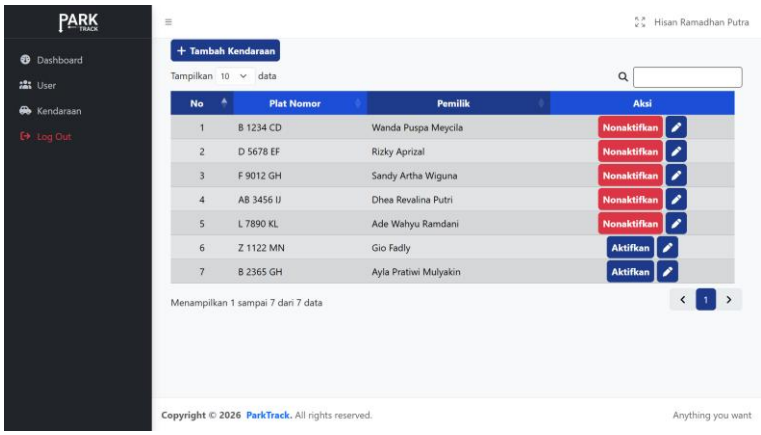

f. Formulir Tambah *User*



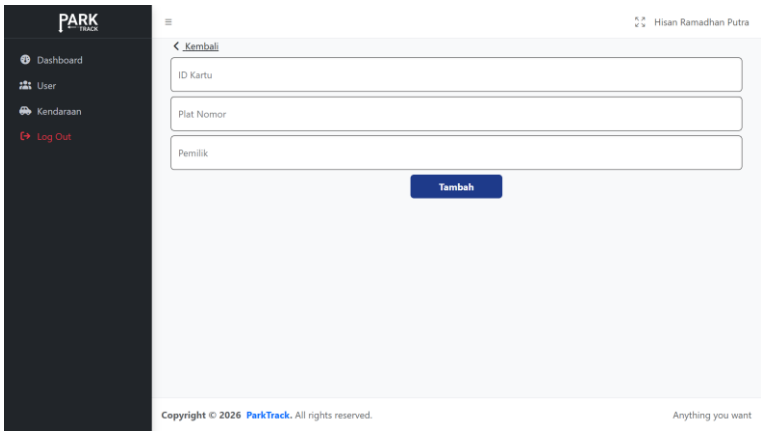

g. Formulir Ubah *User*



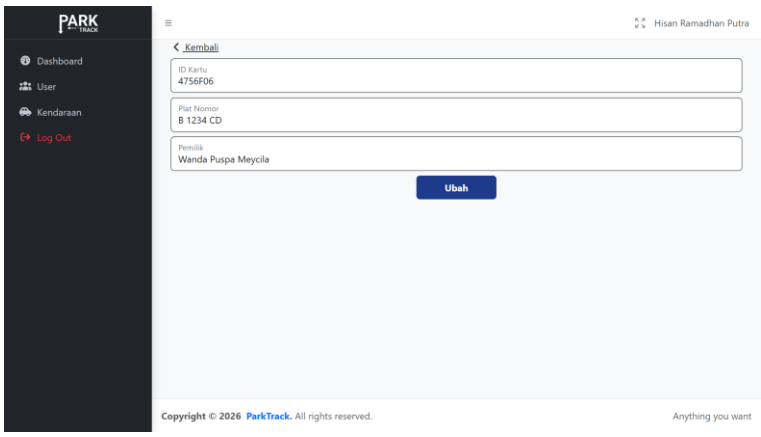
h. Daftar Kendaraan




i. Formulir Tambah Kendaraan

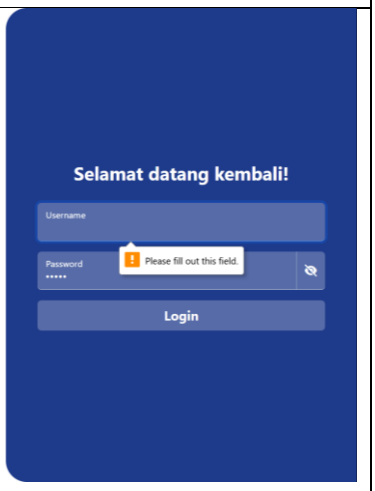
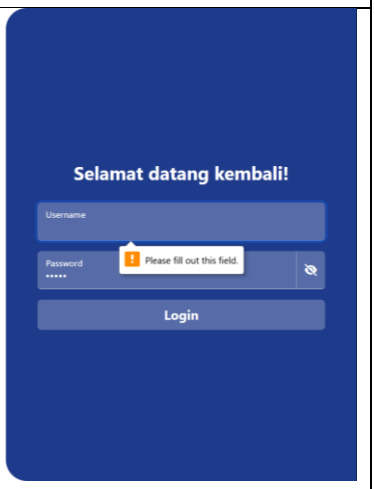
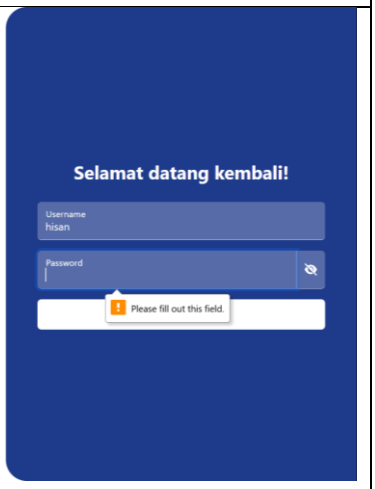


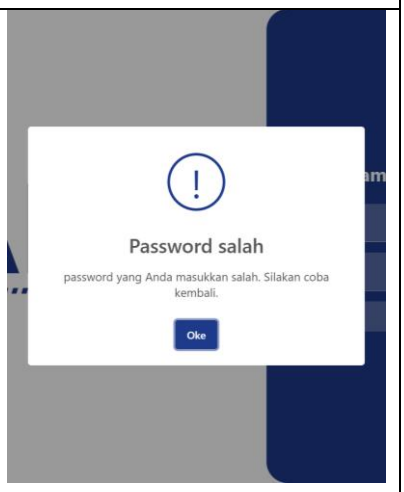
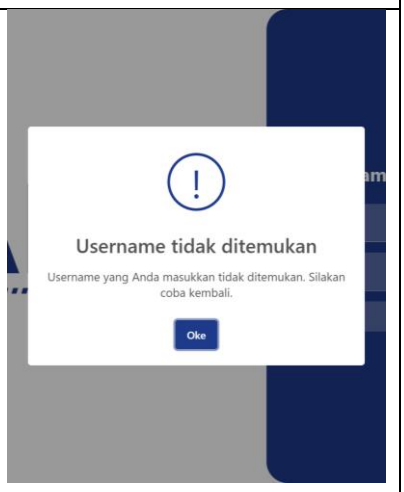
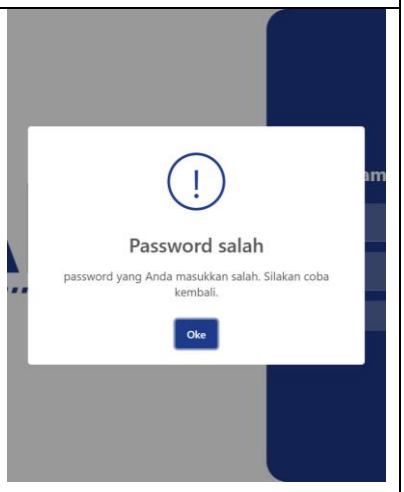

j. Formulir Ubah Kendaraan

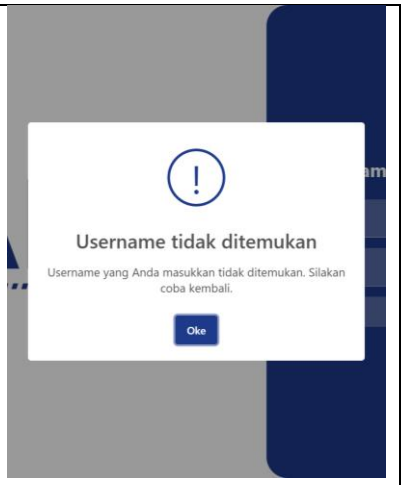
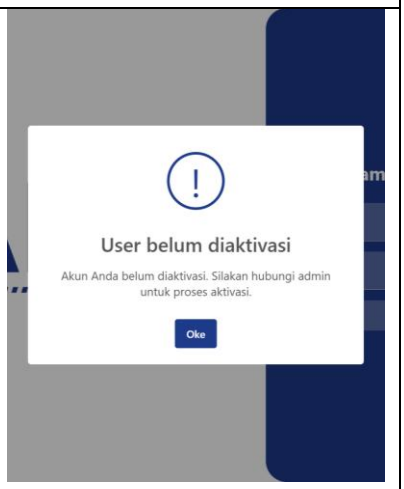


11. PENGUJIAN

a. *Login*

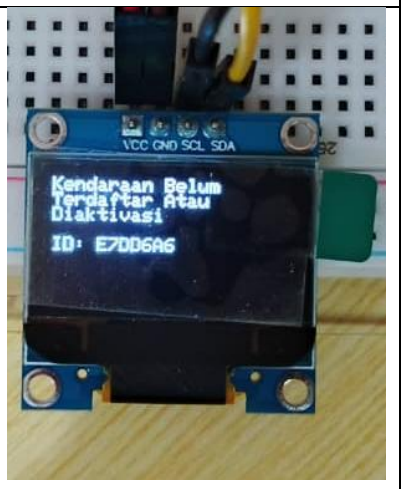
| No | ***Precondition*** | ***Test Case*** | ***Test Scenario*** | ***Expected Result*** | ***Actual Result*** |
|---|---|---|---|---|---|
| 1 | Halaman *login* terbuka | Menekan tombol “Login” tanpa mengisi *username* dan *password* | 1. Tekan tombol “Login” | Sistem menampilkan *tooltip* error *username* dan *password* harus diisi |  <br>**PASS** |
| 2 | Halaman *login* terbuka | Mengisi *password* saja dan menekan tombol “Login” | 1. Isi kolom *password*<br>2. Tekan tombol “Login” | Sistem menampilkan *tooltip* error *username* harus diisi |  <br>**PASS** |
| 3 | Halaman *login* terbuka | Mengisi *username* saja dan menekan tombol “Login” | 1. Isi kolom *username*<br>2. Tekan tombol “Login” | Sistem menampilkan *tooltip* error *password* harus diisi |  |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | **PASS** |
| 4 | Data pengguna terdaftar di *database* | Mengisi *username* dengan benar tetapi *password* dengan salah | 1. Isi kolom *username*<br>2. Isi kolom *password*<br>3. Tekan tombol "Login" | Sistem menampilkan *alert* error *password* salah |  <br>**PASS** |
| 5 | Data pengguna terdaftar di *database* | Mengisi *username* dengan salah tetapi *password* dengan benar | 1. Isi kolom *username*<br>2. Isi kolom *password*<br>3. Tekan tombol "Login" | Sistem menampilkan *alert* error *username* tidak ditemukan |  <br>**PASS** |
| 6 | Data pengguna terdaftar di *database* | Mengisi password tetapi huruf besar/kecil tidak sesuai | 1. Isi kolom *username*<br>2. Isi kolom *password*<br>3. Tekan tombol "Login" | Sistem menampilkan *alert* error *password* salah |  <br>**PASS** |

| | | | | | |
|---|---|---|---|---|---|
| 7 | Halaman *login* terbuka | Mengisi *username* dengan SQL *Injection* | 1. Isi kolom *username*<br>2. Isi kolom *password*<br>3. Tekan tombol "Login" | Sistem menampilkan *alert* error *username* tidak ditemukan |  <br>**PASS** |
| 8 | Data pengguna terdaftar di *database* tetapi berstatus nonaktif | Login dengan pengguna yang belum diaktivasi | 1. Isi kolom *username*<br>2. Isi kolom *password*<br>3. Tekan tombol "Login" | Sistem menampilkan *alert* error *user* diaktivasi |  <br>**PASS** |

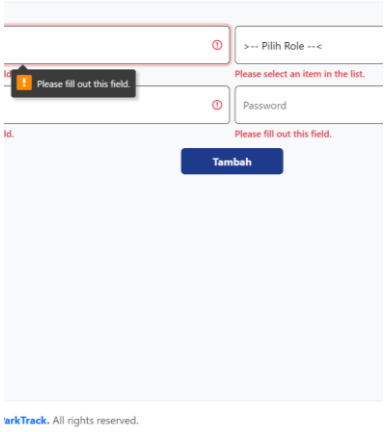
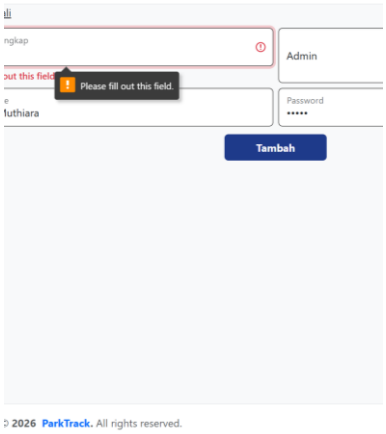
b. Transaksi

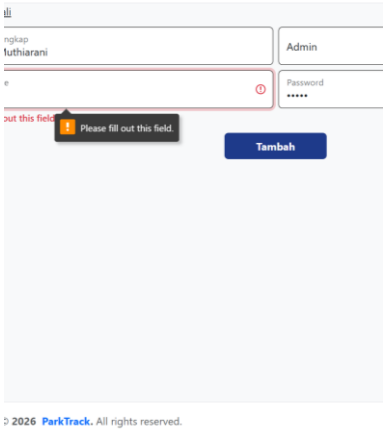
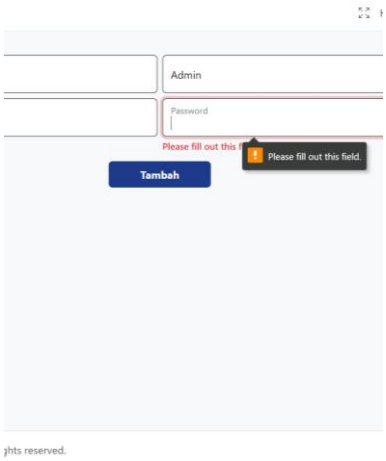
| No | *Precondition* | *Test Case* | *Test Scenario* | *Expected Result* | *Actual Result* |
|---|---|---|---|---|---|
| 1 | Nilai kapasitas dan terisi di *database* sama | Kapasitas area parkir penuh | 1. Pindai kartu | OLED menampilkan pesan kapasitas penuh | |

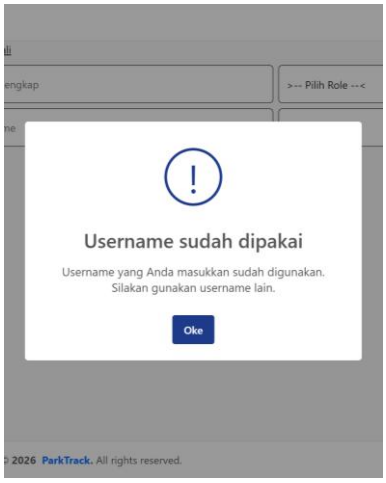
| | | | | | |
|---|---|---|---|---|---|
| | | | | | PASS |
| 2 | Data kendaraan belum terdaftar | Kendaraan yang belum terdaftar mencoba parkir | 1. Pindai kartu | OLED menampilkan pesan kendaraan belum terdaftar | PASS |
| 3 | Data kendaraan belum diaktivasi | Kendaraan yang belum diaktivasi mencoba parkir | 1. Pindai kartu | OLED menampilkan pesan kendaraan belum diaktivasi |  PASS |
| 4 | Data transaksi dengan ID kartu yang sama tetapi berstatus masuk | Kendaraan yang sudah masuk kembali memindai kartu di RFID *entry* | 1. Pindai kartu | OLED menampilkan pesan kendaraan sudah terparkir | PASS |

| | | | | | |
|---|---|---|---|---|---|
| 5 | Data transaksi dengan ID kartu yang sama tetapi berstatus keluar | Kendaraan yang belum masuk memindai kartu di RFID *exit* | 1. Pindai kartu | OLED menampilkan pesan kendaraan belum terparkir | <br>**PASS** |

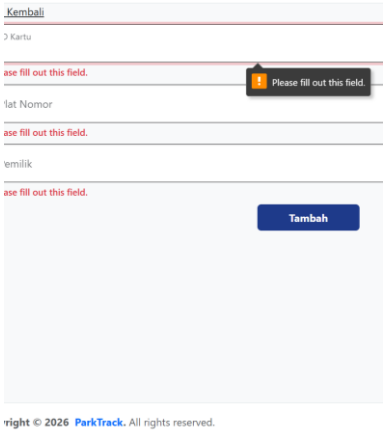
c. Tambah *User*

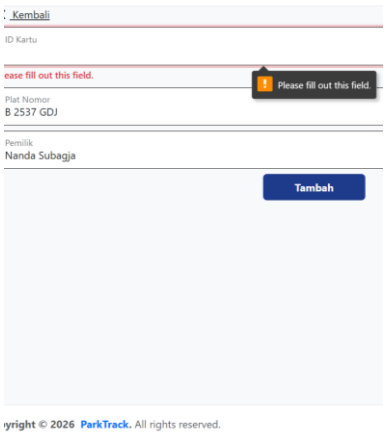
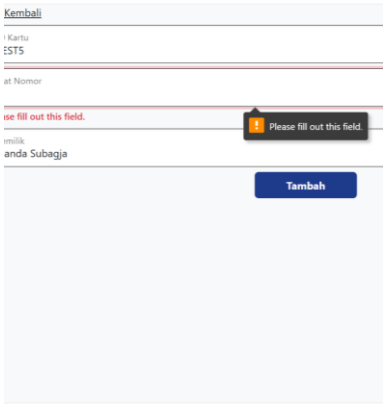
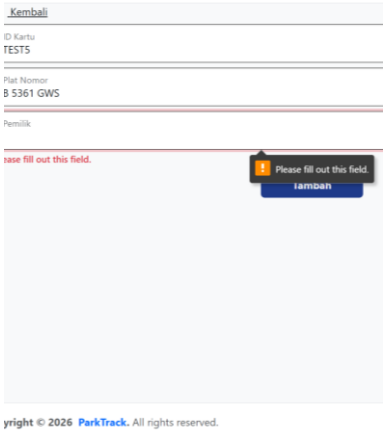
| No | *Precondition* | *Test Case* | *Test Scenario* | *Expected Result* | *Actual Result* |
|---|---|---|---|---|---|
| 1 | Halaman tambah *user* terbuka | Menekan tombol “Tambah” tanpa mengisi *kolom apapun* | 1. Tekan tombol “Tambah” | Sistem menampilkan *tooltip* error semua kolom harus diisi |  <br>**PASS** |
| 2 | Halaman tambah *user* terbuka | Mengisi semua kolom kecuali kolom nama lengkap | 1. Isi kolom *role*<br>2. Isi kolom *username*<br>3. Isi kolom *password*<br>4. Tekan tombol “Tambah” | Sistem menampilkan *tooltip* error nama lengkap harus diisi |  |

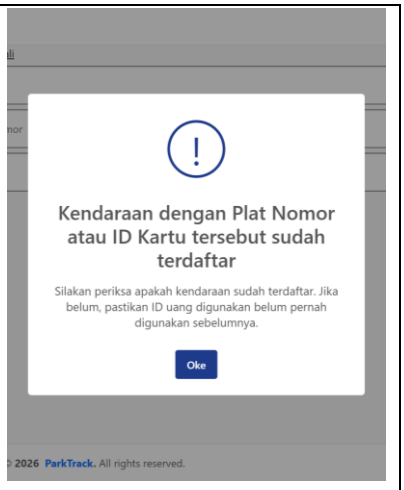
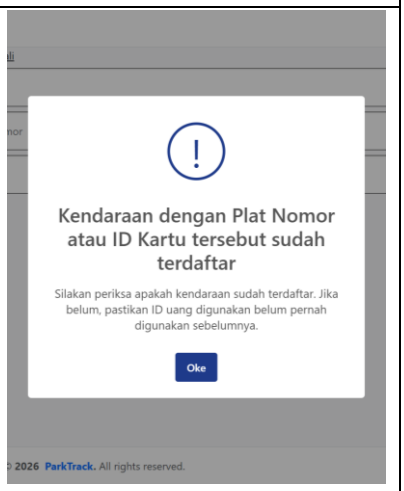
| | | | | | |
|---|---|---|---|---|---|
| | | | | | **PASS** |
| 3 | Halaman tambah *user* terbuka | Mengisi semua kolom kecuali kolom *role* | 1. Isi kolom nama lengkap<br>2. Isi kolom *username*<br>3. Isi kolom *password*<br>2. Tekan tombol “Tambah” | Sistem menampilkan *tooltip* error *role* harus diisi | <br>**PASS** |
| 4 | Halaman tambah *user* terbuka | Mengisi semua kolom kecuali kolom *username* | 1. Isi kolom nama lengkap<br>2. Isi kolom *role*<br>3. Isi kolom *password*<br>4. Tekan tombol “Tambah” | Sistem menampilkan *tooltip* error *username* harus diisi |  <br>**PASS** |
| 5 | Halaman tambah *user* terbuka | Mengisi semua kolom kecuali kolom *password* | 1. Isi kolom nama lengkap<br>2. Isi kolom *role*<br>3. Isi kolom *username*<br>2. Tekan tombol “Tambah” | Sistem menampilkan *tooltip* error *password* harus diisi |  <br>**PASS** |

| | | | | | |
|---|---|---|---|---|---|
| 6 | Sudah ada *username* tertentu di *database* | Menambahkan *user* dengan *username* yang sudah dipakai | 1. Isi kolom nama lengkap<br>2. Isi kolom *role*<br>3. Isi kolom *username*<br>4. Isi kolom *password*<br>5. Tekan tombol "Tambah" | Sistem menampilkan *alert* error *username* sudah dipakai |  <br>**PASS** |

d. Tambah Kendaraan

| No | *Precondition* | *Test Case* | *Test Scenario* | *Expected Result* | *Actual Result* |
|---|---|---|---|---|---|
| 1 | Halaman tambah kendaraan terbuka | Menekan tombol "Tambah" tanpa mengisi *kolom apapun* | 1. Tekan tombol "Tambah" | Sistem menampilkan *tooltip* error semua kolom harus diisi |  <br>**PASS** |

| | | | | | |
|---|---|---|---|---|---|
| 2 | Halaman tambah kendaraan terbuka | Mengisi semua kolom kecuali kolom ID kartu | 1. Isi kolom plat nomor<br>2. Isi kolom pemilik<br>3. Tekan tombol “Tambah” | Sistem menampilkan *tooltip* error ID kartu harus diisi |  <br>**PASS** |
| 3 | Halaman tambah kendaraan terbuka | Mengisi semua kolom kecuali kolom plat nomor | 1. Isi kolom ID kartu<br>2. Isi kolom pemilik<br>3. Tekan tombol “Tambah” | Sistem menampilkan *tooltip* error plat nomor harus diisi |  <br>**PASS** |
| 4 | Halaman tambah kendaraan terbuka | Mengisi semua kolom kecuali kolom pemilik | 1. Isi kolom ID kartu<br>2. Isi kolom plat nomor<br>3. Tekan tombol “Tambah” | Sistem menampilkan *tooltip* error pemilik harus diisi |  <br>**PASS** |

| | | | | | |
|---|---|---|---|---|---|
| 5 | Sudah ada ID kartu tertentu di *database* | Menambahkan kendaraan dengan ID kartu yang sudah dipakai | 1. Isi kolom ID kartu<br>2. Isi kolom plat nomor<br>3. Isi kolom pemilik<br>4. Tekan tombol "Tambah" | Sistem menampilkan *alert* error ID kartu sudah terdaftar |  <br>**PASS** |
| 6 | Sudah ada plat nomor tertentu di *database* | Menambahkan kendaraan dengan plat nomor yang sudah dipakai | 1. Isi kolom ID kartu<br>2. Isi kolom plat nomor<br>3. Isi kolom pemilik<br>4. Tekan tombol "Tambah" | Sistem menampilkan *alert* error plat nomor sudah terdaftar |  <br>**PASS** |

## 12. KODE PROGRAM

a. Controllers/Petugas.php

```
<?php

namespace App\Controllers;

use CodeIgniter\Controller;
use App\Models\Transaksi;
use App\Models\User;
use App\Models\Area;
use App\Models\Kendaraan;
use PhpMqtt\Client\MqttClient;
use PhpMqtt\Client\ConnectionSettings;

class Petugas extends BaseController
{
    public function __construct()
    {
        date_default_timezone_set('Asia/Jakarta');

        $this->transaksi = new Transaksi();
```

```
        $this->user      = new User();
        $this->area      = new Area();
        $this->kendaraan = new Kendaraan();
    }

    public function index()
    {
        $data['area'] = $this->area->first();
        $data['view'] = 'petugas/dashboard';
        return view('petugas/template', $data);
    }

    public function getMasuk()
    {
        $data = $this->transaksi->join('kendaraan', 'transaksi.id_kartu =
kendaraan.id_kartu')->where('status', 'Masuk')->find();
        return $this->response->setJSON($data);
    }

    public function getKeluar()
    {
        $data = $this->transaksi->join('kendaraan', 'transaksi.id_kartu =
kendaraan.id_kartu')->where('status', 'Keluar')->find();
        return $this->response->setJSON($data);
    }

    public function getLog()
    {
        $data = $this->transaksi->join('kendaraan', 'transaksi.id_kartu =
kendaraan.id_kartu')->where('status', 'Selesai')->find();
        return $this->response->setJSON($data);
    }

    public function selesai($idTransaksi)
    {
        $data = [
            'status'  => 'Selesai'
        ];
        $this->transaksi->update($idTransaksi, $data);
        $server   = '127.0.0.1';
        $port     = 1883;
        $clientId = 'ci4-publisher';

        $mqtt = new MqttClient($server, $port, $clientId);
        $connectionSettings = (new ConnectionSettings())-
>setKeepAliveInterval(60);

        $mqtt->connect($connectionSettings, true);
        $mqtt->publish('parking/hisan/lcd', 'Terima Kasih Selamat Jalan',
0);
        $mqtt->publish('parking/hisan/exit/servo', 'TRUE', 0);
        $mqtt->disconnect();
        return redirect()->to('/petugas');
    }

    public function cetakStruk($idTransaksi)
    {
        $data = $this->transaksi->find($idTransaksi);
        $html = view('exports/transaksi', $data);
        $txt = strip_tags($html);
```

```
        return $this->response->download('struk.txt', $txt);
    }
}
```

b. Commands/Subscribe.php

```
<?php

namespace App\Commands;

use CodeIgniter\CLI\BaseCommand;
use PhpMqtt\Client\MqttClient;
use App\Models\Transaksi;
use App\Models\User;
use App\Models\Area;
use App\Models\Kendaraan;

class Subscribe extends BaseCommand
{
    protected $group       = 'MQTT';
    protected $name        = 'mqtt:subscribe';
    protected $description = 'Subscribe MQTT';

    public function __construct()
    {
        date_default_timezone_set('Asia/Jakarta');

        $this->transaksi = new Transaksi();
        $this->user      = new User();
        $this->area      = new Area();
        $this->kendaraan = new Kendaraan();
    }

    public function run(array $params)
    {
        $server   = '127.0.0.1';
        $port     = 1883;
        $clientId = 'ci4-subscriber';

        $mqtt = new MqttClient($server, $port, $clientId);
        $mqtt->connect();

        $mqtt->subscribe('parking/hisan/entry/rfid', function
($topic, $message) use ($mqtt) {
            $cek = $this->area->first();
            if ($cek['terisi'] == $cek['kapasitas']) {
                $mqtt->publish('parking/hisan/lcd',
'Kapasitas Area Sudah Penuh', 1);
            } else {
                $cek = $this->kendaraan->find($message);
                if (!$cek || $cek['status_aktif'] == 0) {
```

```
                    $mqtt->publish('parking/hisan/lcd',
'Kendaraan Belum      Terdaftar Atau       Diaktivasi
ID: ' . $message, 1);
                } else {
                    $cek = $this->transaksi-
>where('id_kartu', $message)->where('status', 'Masuk')-
>first();
                    if ($cek) {
                        $mqtt->publish('parking/hisan/lcd',
'Kendaraan Sudah      Terparkir
ID: ' . $message, 1);
                    } else {
                        $data = [
                            'id_kartu'    => $message,
                            'waktu_masuk' => date('Y-m-d
H:i:s'),
                            'status'      => 'Masuk',
                        ];
                        $this->transaksi->insert($data);
                        $mqtt->publish('parking/hisan/lcd',
'Selamat Datang       Silahkan Masuk', 1);
                        $mqtt-
>publish('parking/hisan/entry/servo', 'TRUE', 1);
                    }
                }
            }
        }, 0);

        $mqtt->subscribe('parking/hisan/exit/rfid', function
($topic, $message) use ($mqtt) {
            $cek = $this->transaksi->where('id_kartu',
$message)->where('status', 'Masuk')->first();
            if (!$cek) {
                $mqtt->publish('parking/hisan/lcd',
'Kendaraan Belum      Terparkir
ID: ' . $message, 1);
            } else {
                $waktuMasuk  = date('Y-m-d H:00:00',
strtotime($cek['waktu_masuk']));
                $waktuKeluar = date('Y-m-d H:00:00');
                $mulai       = strtotime($waktuMasuk);
                $selesai     = strtotime($waktuKeluar);
                $selisih     = ($selesai - $mulai) / 3600;

                if ($selisih < 1) {
                    $selisih = 1;
                }

                $data = [
                    'waktu_keluar' => date('Y-m-d H:i:s'),
                    'durasi'       => $selisih,
```

```
                    'total'        => $selisih * 2000,
                    'status'       => 'keluar'
                ];
                $this->transaksi-
>update($cek['id_transaksi'], $data);

                $terisi = $this->area->find(1);
                $area = [
                    'terisi' => $terisi['terisi'] - 1
                ];
                $this->area->update(1, $area);
                $mqtt->publish('parking/hisan/lcd', 'Total
Biaya:         Rp' . number_format($data['total'], 2, ',',
'.'), 1);
            }
        }, 0);

        $mqtt->loop(true);
    }
}
```

13. REVISI

14. *UPLOAD* KE GITHUB

| | |
|---|---|
| | Repositori bisa juga diakses melalui tautan berikut:<br>https://github.com/Gezaa13/ParkTrack |